

Mathematical Foundation of Computer Graphics

Exercise 4

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In the “code” folder, “GraphCuts” folder is structured as in the exercise’s zip file. “OptimalTransport” folder contains the code for this part as a .ipynb file. I wrote and ran it with the google colab.

“Task2_results” folder contains the image results for the GraphCut’s part.

Task1

I wrote the solution of this part in a .ipynb file with the required results of the images.

Please change the paths of the source and target images if you would like to prove the results.

Task 2

2.1

```
# The result is 19.0 for min-cut value and [False, False, False] for segmentation,
# meaning all nodes belong to the source.

# manual verification: in total 2^3 combinations
# [false, false, false] all three nodes belong to the source, 4+7+8 = 19
# [true, true, true] all three nodes belong to the target, 9+7+5 = 21
# [false, true, true] only node 0 belongs to the source, 4+3+2+7+5 = 21
# [true, false, true] only node 1 belongs to the source, 9+3+2+1+5+7+5 = 32
# [true, true, false] only node 2 belongs to the source, 9+7+1+5+8 = 30
# [true, false, false] only node 0 not belongs to the source, 9+3+2+7+8 = 29
# [false, true, false] only node 1 not belongs to the source, 4+3+2+7+1+5+8 = 30
# [false, false, true] only node 2 not belongs to the source, 4+7+1+5+5 = 22

# [false, false, false] is indeed the mincut segmentation
```

2.2-2.3

You could find the images in the Task2_results folder. What different from the given results in the exercise pdf is the Van Damme segmentation with $\lambda=0.0001$. I don't know if this is usual. The segmentation with $\lambda = 0.002$ is more similar to the reference result in the exercise sheet.

Benefits of graph cut for image segmentation

- Easy user interaction and minimal effort required, rest automatically produced by the algorithm.
- Object and background segments may consist of several isolated connected blobs in the image.
- A globally optimal segmentation can be very efficiently recomputed when the user adds or removes any hard constraints (seeds), no need to recompute from scratch.
- Fast speed in processing 2D and 3D images and videos.
- Applies not only to 2D, but also to N-D images (volumes).
- The seeds can be loosely positioned, stable results regardless of particular seed positioning within the same image object.
- Wide application range, e.g. photo and video editing, medical images and volumes.